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Regulation lite: The rise of emissions trading

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Abstract
Emissions trading is the governmentally promoted hope for a sustainable world. In different contexts, trading regimes display varying potential – both in absolute terms and in comparison with other regulatory instruments. Emissions trading, however, is a device that raises urgent issues regarding its objectives, cost-effectiveness, fairness, transparency, and legitimacy. Its use places emphasis on its "acceptability" and the virtues of regulation that is "lite" because it is non-threatening to the most powerful interests. Emissions trading is resonant with assumptions that are highly contentious – notably that it is acceptable because it involves no losers, or because, in desperate global circumstances, we have no choice but to use it. There is a need to confront the difficult issues presented by emissions trading and to face the challenges of combining "market" and "democratic" systems of legitimization. It is also necessary to avoid taking refuge in all too comfortable beliefs in cumulative checks and balances.

Keywords: emissions trading, legitimacy, regulation.

Introduction

Emissions trading is the regulatory system that is set to eclipse others as the governmentally promoted hope for a sustainable world. The Kyoto Protocol of 1997 established emissions trading as a key instrument in the control of global greenhouse gases and since that date there has followed an explosion of trading regimes and proposals. Most notably, the EU launched its Emissions Trading Scheme in January 2005 and soon afterwards the Stern Review (Stern 2007) advocated the broad use of trading mechanisms to combat climate change.

The essence of emissions trading is that governments decide overall bounds to pollution and then issue a number of permits which confer entitlements to emit pollutants that have a total value equal to the settled upon cumulative limit. Permit holders are then free to buy and sell their allowances in the marketplace (Ellerman 1998; Organisation for Economic Co-operation and Development [OECD] 1999, 2004a,b; Sorrell & Skea 1999; Ellerman *et al.* 2000; Kosobud *et al.* 2000; Norregaard 2000; Butzengeiger *et al.* 2001; National Audit Office 2004; Hansjurgens 2005; Michaelowa & Butzengeiger 2005; European Environment Agency 2006; Tietenberg 2006; Stern 2007).

There is no doubt about the burgeoning popularity and incidence of trading mechanisms, but emissions trading may be a device that can too easily be grasped as a politically convenient panacea. Further scrutiny raises an urgent set of issues regarding

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the objectives that it serves, its efficiency, its fairness and the transparency with which it operates. The rise of emissions trading also introduces new questions concerning our conceptions of good regulation. This article maps out those areas in which emissions trading gives rise to contention and suggests that the current popularity of emissions trading evidences a shift in conceptions of good regulation – away from well-established notions of regulatory merit and toward models that place new emphasis on the notion of "acceptability" and the virtues of regulation that is "lite" in the sense that it is non-threatening to the most powerful interests. This shift, it will be suggested, may demand that we rethink our approaches to regulatory justification and the processes by which we accord legitimacy to regulatory systems.

Part 1 of the article looks at the development of emissions trading systems, the main varieties of emissions trading arrangement, and the claims made for such regimes. Part 2 considers the major challenges and contentious issues presented by emissions trading, and Part 3 makes the argument that the rise of emissions trading may involve a sea change in conceptions of regulation and regulatory justification.

1. The development of emissions trading

The US Acid Rain Program in the US established the first large scale, long-term environmental protection regime to rely on the trading of emissions permits (OECD 1998, 2002, 2004b; Ellerman *et al.* 2000; Environmental Protection Agency [EPA] 2001). The UK Emissions Trading Scheme commenced operating in 2002 and, a few years later, the first international emissions trading mechanism was instituted when the EU Emissions Trading Scheme was launched in January 2005.

Around the world, emissions markets have emerged as the method of choice to price carbon. Norway introduced emissions trading in 2005 for major energy plants and heavy industry, New South Wales operates a scheme for electricity retailers and Japan and South Korea are running pilot programs (Stern 2007, p. 329). Major plans for new emissions trading markets are to be found in the US with the Regional Greenhouse Gas Initiative (RGGI) starting its first compliance period in 2009, and California, Switzerland and Canada all also plan trading schemes (Stern 2007, p. 374). Numerous multinational enterprises have now advocated the use of trading systems in order to control emissions in the most efficient manner, and both Shell and BP have been operating internal trading systems since 2002 (Nicholls 2005).

1.1. The varieties of emissions trading mechanisms

Emissions trading mechanisms have numerous dimensions and there are many varieties of such systems. A basic distinction lies between "cap and trade" and "baseline and credit" approaches (OECD 2002). Under the former, a fixed number of permits are created and each allows the emission of a stipulated amount of pollutant. These permits are allocated or auctioned to firms that are then free to trade them on the open market. In a baseline and credit regime, companies are given performance targets or "baselines" – often set with reference to business as usual (BAU) projections – and they can generate credits by beating their emissions targets. Such credits may then be traded on the open market. With cap and trade, there is a fixed supply of permits for trading, whereas in baseline and credit the supply of credits for trading depends on the regulatees' performance in generating credits by reducing emissions below baselines.

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As for ways to classify trading mechanisms, an authoritative system is one proposed by the US Environmental Protection Agency in 2001 (EPA 2001). This refers to eight basic characteristics (such as scope and method of limiting emissions) and emissions trading mechanisms vary considerably across such properties. Different characteristics bring different strengths and weaknesses (OECD 2001; Butzengeiger *et al.* 2001).[1] Cap and trade systems, for example, are able to fix overall levels of emissions more reliably than baseline and credit approaches. Systems that distribute permits by means of auctions may produce more benefits per unit of emission than processes that "grandfather" permits to established operators (Keohane *et al.* 1997; Boemare & Quirion 2002). Regimes that "retire" percentages of allowances whenever there is trading will produce environmental benefits in a way that non-retirement approaches do not; and different allocations of liability for compliance will affect regulators' capacities to monitor and enforce compliance. Generalizations concerning emissions trading must accordingly be treated with care. It is important to distinguish between the inherent strengths or weaknesses of emissions trading (which are due to the trading process itself) and those contingent matters of performance. The latter flow not from the trading mechanism but from the particular design or characteristics of the scheme (and its fit with the context of application) or the intensity of the policy being furthered, for instance, the stringency of the pollution abatement target that has been set (Harrington *et al.* 2004; Ellerman 2005; Kolstad 2005; Morgenstern 2005; Driesen 2007; Freeman & Kolstad 2007; Tietenberg 2007).

1.2. Why choose emissions trading?

Advocates of emissions trading mechanisms are growing in number and tend to claim a number of virtues for the device (Ackerman & Stewart 1985; Keohane *et al.* 1997; 1998; Stavins 1997; Goulder *et al.* 1999; Keohane 2006; Stern 2007). A first purported virtue is cost-effectiveness. When the trading of permits to pollute is allowed, this means that the burden of reaching a given level of pollution reduction is minimized. Low-cost abaters will have the incentive to reduce pollution levels and sell permits to higher-cost abaters, with the effect that the set level of emissions will be achieved by lowest-cost methods. Unlike flat rate command regimes, market mechanisms tailor abatements to the levels that are cost-effective in each firm or facility. Overall then, compliance with a given limit should cost less than under a command and control regime (Burtraw 1996; Tietenberg 1996; Ellerman *et al.* 2000; Stavins 2002; Ellerman 2005; Harrington & Morgenstern 2007). If intercountry trading is allowed, this ensures that emissions are controlled in the most cost-effective location (Stern 2007, p. 321). Trading also produces rational controls by generating an international price for emissions (Stern 2007, p. 327). Regulatory costs are also said to be low because, once established, the trading system allegedly runs on its own accord.

Another claimed strength of trading is flexibility. Trading gives managers and enterprises the freedom to choose how to deal with their polluting activities. Managers are less restricted than in command regimes because they are at liberty to decide not only the extent of reductions that is cost-effective for their operations but also how to reduce emissions in order to reduce permit costs. They are not restricted by an across-the-board emissions standard or a set of commands that stipulates a particular operational design.

A further posited strength of trading is predictability of outcome. A system established on cap and trade lines is said to offer far more certain outcomes than, say, a

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taxation regime. Overall levels of emissions are fixed in a cap and trade regime but, in a taxation system, they are contingent on individual firms' cumulative responses to incentives. This predictability is of value in restraining pollution levels below important threshold points (e.g. levels at which wildlife will be killed) and in calculating such matters as compliance with Kyoto Protocol undertakings (Faure *et al.* 2003).

An emissions trading regime is also said to have the potential to deal with distributional issues. It can control the assignment of emissions through the choice of initial allocation method for permits (Ackerman & Stewart 1985; Stavins 1997; Stern 2007).

An emissions trading system, it is further claimed, stimulates innovation in the techniques and technologies of emissions control (Ackerman & Stewart 1985; Keohane *et al.* 1997; Stavins 1997; Goulder *et al.* 1999; Keohane 2006). It does so since a firm that finds a cheaper way to abate can then generate allowances to sell on the market. Such firms, accordingly, will look to the producers of new techniques and technologies to provide them with novel ways to abate. Those producers will, in turn, have an incentive to invest in the appropriate research and development regarding abatement strategies.

Emissions trading mechanisms are said to handle competition and coordination well. On the international stage, for instance, emissions trading schemes can produce a common price (e.g. for carbon) across countries and can do so more easily than processes involving the harmonizing of taxes. Trading schemes can thus, it is said, introduce carbon pricing without competitiveness implications between participating countries. It is thus a "very powerful tool in the framework for addressing climate change at an international level" (Stern 2007, p. 327).

Finally, emissions trading is claimed to produce political advantages. Trading mechanisms offer a means of introducing controls but also of avoiding major opposition from entrenched incumbents. They can do so by grandfathering allowances. Even when permits are allocated by auction, trading allows established actors to defend their positions by exploiting their accumulated wealth. When international regimes are at issue (for example, in the wake of Kyoto), a trading system will involve transfers from developed to developing countries (or their enterprises) and this encourages acceptance of the international regime. Advocates of emissions trading also argue that the device enhances the democratic quality of the policy-making process. It does so, they say, because it focuses discussions clearly on the key issue of the overall level of pollution that should be established – which contrasts with the opacity of command and control systems that center debates on arcane questions of best available technology versus other formulations of standards (Ackerman & Stewart 1987).

2. Challenges and issues

Despite their potential advantages, emissions trading mechanisms raise a series of contentious issues. They are not free from difficulties, and a brief review of emissions trading's main alleged weaknesses and areas of contention will show how much emissions trading still needs to be justified. Such a review also provides a useful starting point for considering how the rise of emissions trading evidences a shift in both the way governments and others see the role of regulation and the way that regulation may be legitimized.

A preliminary caveat should, however, be entered. As already noted, emissions trading regimes come in widely differing forms and are applied in divergent contexts, which

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makes generalizing a fraught process. In reviewing potential issues of contention, it should be emphasized both that some trading systems – for example, the US Acid Rain Program – have been heralded as considerable successes (Ellerman *et al.* 2000) and that other policy instruments such as command regimes are by no means problem free (Harrington *et al.* 2004).

2.1. Targeting and objectives

A first issue with emissions trading concerns the objectives to be pursued. A trading process, in itself, offers no benefit to, say, the environment. It does not reduce greenhouse gas emissions. What it does do is to provide a way for a given target to be achieved at lowest cost. In a cap and trade system, it is the setting of the cap that provides the opportunity for imposing limits on, say, environmental pollution. Within a baseline and credit regime, it is the setting of the baseline. Proponents of emissions trading would, however, argue that emissions trading systems offer highly implementable ways of reaching whatever targets are decided upon and that they lend themselves to strategies for tightening controls as trading takes place (Stern 2007, ch. 15).

It should be noted here that, in meeting targets with a trading device, much depends on the mode of defining emissions and distributing allowances. If emissions are defined absolutely (i.e. an absolute limit to discharges is set) this targets environmental objectives directly. If, however, emissions are defined relatively (as limits per unit of production), increases in levels of productivity may produce overall increases in emissions even when there is full compliance. (The same would be true of a command system imposing design standards.)

Are emissions trading systems amenable to the institution of environment enhancing targets? Experience with the European Union Emissions Trading System (EU ETS) points to a set of potential challenges to be faced – and, notably, to the difficulties that are experienced when allocating not through auctions but by means of governmentally established entitlements (Convery & Redmond 2007; Ellermam & Buchner 2007; Kruger *et al.* 2007). Crucial in the EU ETS were the initial allocations or "allowances" for Phase 1 of the regime in 2005–2007. Incumbent enterprises were extremely concerned to generate generous allowances that would minimize any potential costs, and the emissions trading directive left it to Member States to establish allocation plans. As a result, intense lobbying ensued across the EU so that: "[I]n most cases, these efforts resulted in lax emissions targets, complex special allocations to powerful interest groups and, in some cases, even in an over allocation compared to actual emissions" (Butzengeiger & Michaelowa 2004, p. 118; Svendsen 2005; Open Europe 2006; Convery & Redmond 2007).[2]

Additionally, implementation timescales were tight in the EU ETS and a complex set of allocation rules had to be worked to. As a result, it can be argued, powerful interests were able to exploit their informational advantages to keep the constraining effects of the ETS at bay (House of Commons Environmental Audit Committee 2006–2007). For example, Greenpeace protested:

> Governments massively over allocated CO2 permits as the market crash in the carbon price has shown ….(the price fell by more than 60%)….it was because the system relies on future emissions projections as a method to set a cap and then gives out permits for free. Industry simply inflates its own emissions projections in order to ensure it maximises the number of free permits that it gets – permits that, once allocated, have a significant market value. (Oakley 2006)

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The pressure group dubbed this "a license for polluters to print money" and the German environment minister reported that the EU's four largest power producers had profiteered from the ETS at the expense of consumers – and had stoked their earnings by €6bn–8bn (IPA Consulting 2005; Gow 2006; Open Europe 2006).

When the UK revised its EU ETS National Allocation Plan for the post-2007 period, its proposed cap on emissions for Phase 2 of the ETS was higher than for Phase 1, prompting environmental pressure groups to accuse the UK government of repeating its over-allocation of permits to the detriment of consumers and the environment (Greenpeace 2006).

The EU experience raises doubts about the amenability of some emissions trading systems to the progressive adjustment of targets in order to improve environmental protections. Command systems have been criticized on this front (Ackerman & Stewart 1987, p. 174) but it can be argued that emissions trading regimes are similarly beset by difficulties of complexity, uncertainty, and delay when approaching the revision of standards and limits (Driesen 1998a, 2003). There will also be lobbying difficulties and pressures from potential litigants; these are likely to prove to be at least as severe as those encountered in traditional command regimes.

The message to be drawn from the EU ETS is that if allowances are distributed at no cost, there are serious incentives to distort emissions projections so as to create windfalls. One answer to this problem is to allocate allowances by means of auctions. If auctions are competitive, polluting enterprises will calculate their abatement costs as accurately as they can and then: (i) take steps to abate where this is cheaper than purchasing permits; and (ii) purchase permits to cover production up to the point of non-profitability. Many "grandfathered" firms will, of course, object to having to pay for emissions that previously had been discharged at no cost. Auctioning, however, can be defended on the grounds that, not only does it avoid dangers of manipulation, but that polluters and the consumers of polluting products should have to pay for the harms that they inflict on the environment.

2.2. Is emissions trading cost-effective – does it produce lowest-cost abatement?

2.2.1. Innovation and technology forcing

Proponents of emissions trading schemes claim, as noted above, that such mechanisms can usefully drive forward the search for more cost-effective abatement technologies as traders seek to lower costs (Ackerman & Stewart 1985; Stavins 1997; Ellerman *et al.* 2000; Keohane 2006). Emissions traders, it is contended, will look to purchase abatement technologies when the costs of abatement per unit are rendered less than the costs of permits. Skeptics, however, argue that such incentives have limitations and that there are solid reasons to suspect that an emissions trading program does a poorer job of stimulating innovation than a comparably designed traditional regulation (Driesen 1998a, p. 325, 2003, 2007; Fischer 2005). A central argument is that in a cap and trade regime, the cap sets a limit to emissions and the effect of trading is to concentrate emissions reducing efforts on those facilities that have the lowest abatement costs – which will tend to be the operators with low-tech systems. Emissions trading, it is said, compares poorly with traditional regulation regarding high-end innovation. This is because trading reduces the incentives for high-cost facilities to innovate in order to save costs (Malueg 1987): "Why bother making expensive long term structural changes if you can meet your

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pollution rights from operators that can cut their carbon cheaply?" (Lohmann 2006, p. 18). In short, high-end buyers of credits will tend not to innovate but low-end sellers will seek to release credits by implementing low-tech changes. One response might be that the scenario described is cost-effective, but such a response fails to account for the syndrome of falling abatement costs – that is, the propensity of initially expensive innovations to develop in the longer term (notably through economies of scale) into low-cost abatement mechanisms. The effect of emissions trading, at least under certain conditions, may be to postpone or reduce the chances of discovering innovative abatement systems (OECD 2002; Driesen 2003).

Short-term gains may tend to be purchased in emissions trading regimes at the cost of the development, in the longer term, of new technologies that may revolutionize environmental performance. To give an example: within an emissions trading scheme the fossil-fuel burning electricity utility in the developed country may claim credits for activities undertaken abroad as a substitute for reducing greenhouse gases at home. Had the utility been faced with a command requirement, it might have been stimulated to take more radical steps such as considering changing fuels or employing alternative technologies such as innovative fuel cells or solar energy solutions (Driesen 2003).

Will emissions trading systems encourage the closure of old polluting premises and the building of innovative, high-tech and low polluting establishments? If a company is considering renewing its plant, the opportunity to sell off its released allowances will create an incentive to innovate. If, however, governments seek to tighten overall caps by withdrawing allocations when installations are closed, this creates perverse incentives to keep old inefficient units operating in order to preserve the value of those allocations. Similarly, if governments attempt to encourage new enterprises by giving free allowances to new entrants, this creates perverse incentives to maximize such allowances by building highly polluting systems (Carbon Trust 2006, p. 8). When grandfathering is used to allocate permits at the inception of a trading regime, this will reduce the value of permits and reduce incentives to innovate (Milliman & Prince 1989; Boemare & Quirion 2002).

2.2.2. Uncertainty

A further potential difficulty with emissions trading systems is said to be uncertainty (Stern 2007). In the first place, if firms are familiar with command regimes and are not certain about the longevity of an emissions trading scheme they will not be inclined to make cost-effective strategic decisions (Stavins 2005, pp. 67–68). In a baseline and credit system, excessively generous initial allocations of allowances can produce volatilities in the price of emissions (World Bank 2007). Parties considering investing in research into abatement technologies may be discouraged by such volatility and the attendant financial uncertainties (Carbon Trust 2006, p. 8). Firms may respond to uncertainty by adopting a "wait and see" approach' (Ben-David *et al.* 2000, Blyth & Sullivan 2006). Carbon prices in the EU ETS have proved volatile (Convery & Redmond 2007) and this has allegedly discouraged meaningful investment in carbon reducing technologies and encouraged short term trading of positions to optimize returns and limit risks (Open Europe 2006). In any regime where there is a cap on permits in supply, small changes in demand can lead to large changes in prices. This volatility is particularly damaging in industries such as electricity generation where investment decisions work to long horizons (Lockwood 2007). When cutbacks in emissions caps are set with reference to BAU projections, this process involves particularly high levels of uncertainty since the cutbacks are made "from

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moving targets" (House of Commons Environmental Audit Committee 2007). Command regimes will involve uncertainties regarding the stringencies of the governmentally imposed requirements of the future but emissions trading systems can create uncertainties in relation to both the limits that governments or regulators set on caps and the emissions prices that are established in the marketplace (Driesen 2003; Dennis 1993).

2.2.3. The health of markets

Another major issue in emissions trading is the state of the market. If there is no vigor in the trading of permits, there cannot be a strong set of incentives that will influence abatement behavior. The parties within the system have to be both disposed to trade and able to do so if emissions trading is to operate as an effective control mechanism. Some emissions trading systems have been said to involve healthy markets (Ellerman *et al.* 2000) but, in others, trading activities have been far lower than expected – and have sometimes been zero (Swift 1997; Kraemer *et al.* 1998; OECD 2002).

According to the OECD, experience demonstrates that emissions trading programs can take a long time to develop (OECD 2002; Stavins 2005). A further difficulty is that if initial allocations of permits are defective and there is governmental correction ex post, this may create further uncertainties in the system as the rules change (Carbon Trust 2006).

2.2.4. Transaction costs

Some emissions trading schemes have gained reputations for low administrative costs (Ellerman *et al.* 2000; Tietenberg 2007, p. 70) but others are complex and, particularly when targeted "downstream" (i.e. toward consumers in the supply chain), will tend to raise difficult issues regarding administrative and transaction costs (Stavins 1995; Sterner & Hammar 2005). Thus, when the OECD considered controlling pollution by using tradable permits to ration mobility in the transport sector, it noted the administrative costs of targeting a large number of mobile sources and the high transaction costs involved in making permits transferable. It concluded that fuel taxes would be a cheaper solution (OECD 2002, p. 148). To return to the EU ETS, this has been dubbed "an administrative nightmare" whose complexities impose huge burdens of an estimated £62m on firms and public sector bodies – burdens that are said to be felt especially by small plants that are covered by the scheme but contribute little to emissions (Open Europe 2006).

2.2.5. Information

A cost-effective trading system will be one that is based on reliable data and within which there are good information flows (Ellerman *et al.* 2000). On this front, an issue with such systems, and notably with baseline and credit approaches, can be their vulnerability to data manipulation and, when allowances are issued at no cost, the encouragement of such manipulation (Berland *et al.* 2001; Open Europe 2006; Ellerman & Buchner 2007). Such difficulties present particular problems when the success or failure of the system is sought to be assessed (Ellerman & Buchner 2007). A familiar criticism of traditional "command" regulation is that regulated firms are able to exploit the information asymmetry between regulator and regulated. It is arguable, though, that in some respects the incentive to manipulate may be worse in an emissions trading mechanism. That incentive is, for instance, linked to the producing of a firm-specific gain – one that can be expected to bring competitive advantages – rather than to an impact on the stringency of a command standard that will be applied across the board to all relevant firms.[3]

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2.2.6. Enforcement

It should be emphasized that emissions trading markets will generally need regulatory encouragement if they are to develop (Harrington & Morgenstern 2007). The rules of trading must be monitored and enforced since non-observance of allowances will undermine the value of trading and negate ceilings on emissions (Tietenberg 2006; Stern 2007, p. 336). On the international front, confidence in trading systems demands not merely that compliance systems are strong across participating nations but that there is agreement on standards for the monitoring, reporting, and verification of emissions.

Emissions trading schemes, accordingly, do not escape the enforcement challenges that are familiar in command regimes (Greenspan Bell 2002; Kruger *et al.* 2002). Under some conditions, trading schemes have proved conducive to high levels of compliance (Ellerman *et al.* 2000; Harrington *et al.* 2004). Under other conditions, however, it is arguable that emissions trading systems may render enforcement particularly difficult. Within an international greenhouse gas trading regime, for instance, an enterprise within a developed country will look to buy allocations as cheaply as possible. The very lowest prices are likely to be those offered by firms in developing countries whose governments are poorest at monitoring and enforcing (Richman 2003). This will be the case because the selling firms will anticipate that, thanks to poor enforcement, they can sell their allowances but still carry on emitting at the usual levels.

Cross-jurisdictional emissions trading systems, indeed, may encourage tokenistic enforcement and even corruption, since unethical members of governments will often be able to reap personal gains and, at the same time, offer home enterprises competitive advantages. Emissions trading can be said to place heavy stress on enforcement but to involve enforcement under extremely difficult conditions. A special concern in the global warming context may be that emissions trading mechanisms involve huge verification challenges and create the dangerous illusion that production patterns in the North can be maintained without harming the climate.

2.3. Is emissions trading fair?

A fundamental problem with market-based systems of distribution is that such systems have an inherent bias in favor of those parties who possess wealth and they tend to remove power from those who lack resources. The results of trading may be claimed to be cost-effective but this does not ensure fairness (OECD 2002, p. 20).

A first difficulty with trading systems is that, if they are to overcome the political hurdles of inception, they tend to have to "grandfather" existing operators into the system (OECD 1998, p. 39; Raymond 2003). If, however, permits to pollute are allocated on the basis of historical or current emission levels, polluters will not "pay". They will be rewarded for their records of pollution and will be positioned so as to be able to maximize their rewards by exploiting their informational advantages and abilities to manipulate data to their advantage (Baer *et al.* 2000; Neuhoff *et al.* 2006; Open Europe 2006; Stern 2007, p. 333).[4]

Free allocations, moreover, may result in windfall profits. As Stern commented, "Not surprisingly, free permits are generally favoured by existing players in industry" (Stern 2007, p. 333; von Malmborg & Strachan 2005).

Fairness, Stern added, demands that historical polluters are not simply rewarded but should pay a greater share of abatement costs (Stern 2007, p. 472). The difficulty with this argument is that such redistributive approaches always tend to be countered by the

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regressive effects of trading systems. As for comparisons with other regulatory methods, such as command and control, it has been argued that most empirical studies find that, across a range of policy instruments, the costs of control tend to be borne disproportionately by poorer groups – but that this is especially the case with grandfathered emissions permits (Parry 2004; Parry *et al.* 2005).

Nor do fairness issues disappear if permits are allocated by auctioning rather than by free allocation. Auctioning favors those incumbents who have the existing resources to make successful bids. The principled objection here is that it is unfair that incumbent polluters – who have accumulated wealth at the cost of the environment – should be better positioned than non-polluters or new entrants to the field.

Such unfairness can result in barriers to market entry (Hahn 1984) and small and medium enterprises may also complain that they suffer competitively because they are far less able than large companies to deal with the extensive administrative and informational burdens that are involved in negotiating allowances or organizing bids for permits (Butzengeiger & Michaelowa 2004). On the international stage, it has similarly been argued that: "Only big firms can afford to hire carbon accountants, liaise with officials and pay the costs of getting projects registered with the UN. Yet these are often the companies that local people battle hardest against in defense of their livelihoods and health" (Lohmann 2006).

Post Kyoto, a key issue is the development effect of trading systems (Richman 2003). Internationally, emissions trading solutions have been said to involve a double injustice. The effects of existing emissions are felt disproportionately by the less developed nations and they restrict development over coming years. Trading has been called "colonialism with a modern face" insofar as it perpetuates and deepens inequalities of access to and control of resources (Rising Tide 2006). Critics protest that trading allows wealthy countries and companies to escape their historical responsibilities for greenhouse gases, to avoid making emissions reductions in their own operations, and to "defraud developing countries of their rights to use of the global atmosphere" (Christian Aid 2002).

The charge, then, is that historically based allocations allow currently high emitters to impose environmental damage on other countries and to lock the less developed nations into lower levels of development. The linked concern is that in the early years of trading, the mechanism allows existing industrialized users to meet their targets at lowest cost and to avoid making reductions in home emissions. When, however, developing countries become faced with emissions targets themselves, the cheapest forms of emissions abatement will have been exhausted and only more expensive high-tech forms will be left – at which time industrialized countries will be unwilling to invest abroad. In short, industrialized countries will have gained preferential use of lowest cost abatement methods and reaped a competitive advantage while suppressing development (Christian Aid 2002, p. 7; Richman 2003). Supporters of emissions trading might argue that such considerations can be taken into account when allocations are negotiated; but this response makes assumptions about the bargaining power and positions of developing countries (or the altruism of developed countries) that may be unrealistic – a matter to be returned to below.

A central concern regarding global fairness is that developing countries cannot reasonably be expected to restrict their future emissions without being assured of a fair allocation scheme that will not impair their ability to develop (Aggarwal & Narain 1991; Baer *et al.* 2000). This demands, it can be said, not historically based or auction based

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distributions but allocations based on equal rights to the atmospheric commons for every individual.

A further argument, however, suggests that, from a development point of view, it is not enough to allocate emissions rights on a per capita equal rights basis. The effect of this would be to allow existing wealthy polluters to purchase, from poor permit holders, sufficient allocations to allow them to continue to trade at profit maximizing levels. There would be a one-off transfer of wealth to poorer firms but these less wealthy players would be paying a price for that transfer – in the form of forfeited opportunities (Mehmet 1999; Richman 2003, pp. 149–154).

Informational asymmetries would be likely to exaggerate this effect (Gupta 1997; Richman 2003). To take an example, let us suppose that it is decided internationally to cap pollution from air travel and to do so by establishing a trading scheme in which all companies are allocated x hours of flights per year (size of allocations to reflect numbers of employees). Wealthy Company A, from a developed country, would, say, purchase the emissions allowances of less developed companies B, C, and D. Would the price paid reflect the true wealth generating potential of those allowances? It is unlikely to do so because not only has Company A a greater capacity to develop that potential (which is what makes the system efficient) but it has superior information about that potential.[5] Companies B, C, and D, moreover, are likely to suffer from non-informational factors that will further undermine their abilities to strike satisfactory deals with Company A. Notably they are likely, if sited in a developing country, to be competing, as sellers of allocations, with firms that are less well informed, less rational, and more desperate to sell. The overall effect of allocations trading on Companies B, C, and D is that they receive a one-off payment (a suboptimal one) and, being excluded from air travel, they will have restricted development potential and are likely to be left ever further behind in the marketplace by Company A. The propensity of companies B, C, and D to opt for the short-term profit at the expense of the longer-term gain is, furthermore, consistent with the message from the risk literature that actors tend to discount the future effects of their actions (Cropper & Portney 1990; Viscusi 1992; Graham & Weiner 1997).

According to Stern, one of the major advantages of emissions trading systems is that they allow efficiency and equity to be considered separately (Stern 2007, p. 473). The UN Framework Convention on Climate Change (UNFCCC) approaches this issue and argues that developed countries should show leadership in tackling emissions, transferring technology, supporting capacity building, and financing the incremental costs of emissions reductions.

These may be sentiments worthy of support but we should be clear about the degree to which emissions trading and reallocative policies pull in opposing directions (Richman 2003). Such a tension may be so severe as to lead cost-effectiveness concerns to swamp those of equity – which negates Stern's argument that emissions trading conveniently allows equity and efficiency issues to be considered separately. Thus, it has been argued that: "Emissions trading may conflict with the post-Rio developed country leadership principle in several ways. Most obviously, it allows developed countries to claim that they are meeting their reductions obligations through trading and to 'double count' trades as both domestic reductions and assistance to developing countries" (Richman 2003, p. 170).

Emissions trading exaggerates the effects of inequalities in wealth distribution and offers up wealth creating opportunities to the currently wealthy (and often polluting).

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Reallocative policies, when linked to emissions trading, may look transparent and worthwhile, but three points are worth stressing. First, any reallocative virtues will be due to distributional decisions and restrictions that are placed on the trading mechanism – not to the trading mechanism itself. Second, any protections for the less well off, less powerful, less developed and less well informed will be operating within a system that is intrinsically skewed in favor of wealth holders. Finally, it can be argued that, as far as fairness is concerned, there are grounds for doubting whether emissions trading systems match up to the performance of command or taxation regimes. The latter, after all, offer across-the-board controls, are generally more easily enforced from the center and are not so vulnerable to distortion in favor of the well resourced.

2.4. Is emissions trading accountable and transparent?

It has been argued, as noted above, that emissions trading combines democratic accountability with a market mechanism and that trading focuses public attention on decisions about aggregate emissions reductions (Ackerman & Stewart 1987). In this regard, it is claimed that emissions trading can offer more democratic accountability than the rule-making processes of traditional command regulation. Skeptics, however, argue that trading systems have a special complexity that does not facilitate access. Such systems, it is complained, overlay market processes on top of the standard setting procedures usual to command regimes. This duality, it is said, makes citizen participation in emissions trading programs more difficult than in traditional regulation and renders such programs highly vulnerable to industry lobbying (Driesen 1998b; Lockwood 2007, p. 7).

Special criticisms may apply to systems, such as the EU ETS, in which caps are set in relation to BAU projections. As noted above, such approaches mean that the caps imposed on emissions are liable to change as firms change their forecasts of emissions – a process that has been said to create an obvious lack of transparency that underlines the need to set reductions from an absolute level of emissions, rather than from a baseline of BAU projections which may vary significantly according to the differing assumptions that are fed into them (House of Commons Environmental Audit Committee 2007).

Accountability to whom is, of course, also a key issue, and one of the recurring criticisms of carbon trading post Kyoto is that it makes policy-makers responsive to multinational corporations, not local populations (Wall 2006). The emissions trading device, moreover, involves a lack of accountability by public officials for the distributional decisions of the market in allocations – regarding, for instance, the location of the steps that are taken to abate emissions or the competition consequences of allocations. The emissions trading process, as a result, helps such officials to avoid specificity about the policies being furthered through the trading mechanism and the distributions of costs and benefits, winners and losers (von Malmborg & Strachan 2005). If, moreover, trading is allowed across jurisdictions, there may be the additional problem of perverse incentives. The purchasers of permits may be induced, by emissions trading, to purchase credits from countries that monitor credit generating activity poorly. Monitoring in such countries will tend to be particularly weak if the pollution at issue is not inflicted on that country specifically but is spread across nations as a "common bad" – as with greenhouse gases (Mabey *et al.* 1977, p. 25; Driesen 1998a, p. 15). Such weak monitoring systems will undermine accountability, and transparency will be especially poor.

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How, then, does emissions trading score on general transparency? One way to summarize this issue is to evaluate emissions trading processes with reference to Stirton and Lodge's four key transparency mechanisms (information, choice, representation, and voice) (Stirton & Lodge 2001). Information allows informed choices by consumers and others but, as noted, emissions trading supplies little information to the consumers of products regarding the emissions abating efforts of suppliers and manufacturers or the locations at which any abatement efforts are being made. Choice allows consumers to choose the nature of products and goods – but, again, the lack of information provided to consumers in emissions trading systems means that purchasers of products are ill-placed to choose between polluting and non-polluting products. Representation ensures transparency by allowing access into policy processes to user and interest groups. With emissions trading mechanisms, however, such access is conferred predominantly on those suppliers who buy and sell permits; other groupings are, in the absence of disclosure regulation, kept at a distance by their non-inclusion in the market. Voice allows user participation and redress. Unfortunately, however, the consumer of goods has no access to the trading market, and the processes used to set caps and baselines tend to be dominated by conversations between supply firms and governments. Overall, then, serious doubts arise concerning the general transparency of emissions trading processes – at least to ordinary consumers.

3. The new regulation lite

3.1. Emissions trading and regulatory philosophies

The philosophical significance of emissions trading lies in its shifting the implicit measures of regulatory quality and legitimacy. Since the late 1990s in the UK, for example, regulation has been assessed, across government, with respect to the Five Principles of Good Regulation and the yardsticks of Proportionality, Accountability, Consistency, Transparency and Targeting (Better Regulation Task Force [BRTF] 2003). New concerns to reduce regulatory burdens (BRTF 2004, 2005; Gershon 2004; Hampton 2005) have set the search for good regulation at tension with the quest for less regulation and have evidenced some philosophical confusion within government (Baldwin 2005, 2007). The rise of emissions trading has taken place amid that confusion and, it is contended here, has proven consistent with both newly thin notions of legitimacy and a weakening of expectations that good regulation should satisfy the Five Principles. Two separate but powerful assumptions can be seen, in turn, as consistent with such changed approaches to legitimacy and regulatory evaluation.

The first of these assumptions is that the emissions trading device is a relatively uncontentious method of controlling emissions cost-effectively and accountably while reducing private sector compliance costs (Ackerman & Stewart 1987). Driesen (1998a) argues that proponents of emissions trading employ a "free lunch" metaphor in presenting such an argument. This is a metaphor that suggests that there are no real losers in distributing pollution rights by auctions or other allocation methods and then allowing trading. Efficiency is held to be the winner and rights will flow to those parties who will maximize the value that can be extracted from the relevant emissions causing processes. In defense of the Stern Review, it might be noted that its authors did accept that issues of distributional and social justice, as well as development, arise when emissions trading devices are used. Once the existence of such issues is acknowledged, however, the

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advocates of trading tend to deal with the solutions to such thorny problems in a manner consistent with the "free lunch" metaphor. Thus, emissions trading is lauded as possessing the virtue of bringing distributional issues out into the daylight and little is made of its built-in tendency to exacerbate wealth and development inequalities. Emissions trading is portrayed as fair and progressive in effect.[6] This picture flows from the placing of emphasis on short-term gains to less well off parties and the underplaying of the longer-term losses that trading will impose on those same actors (Dresner & Ekins 2004; Roberts & Thumim 2006, p. 5). This portrayal is exemplified in the *Guardian* writer Polly Toynbee's analysis of trading:

> Miliband's electric radicalism comes in his plan for personal carbon allowances. Here is where social justice meets green politics for the first time. Give every citizen the same quota of energy and let them buy and sell it on the open market. The half of the population who don't fly will make money from selling their quota to the half who do. Drive a gas-guzzling 4x4 and you will have to buy a quota from the third of the population with no access to a car. Who could complain about such transparent fairness? …….[I]t in effect redistributes money from the rich to the poor. (Toynbee 2006)

Respondents to such an approach would, however, be sure to complain that the commodification of rights to emit carbon constitutes not "transparent fairness" but a means of magnifying wealth differentiations and of pricing poorer citizens out of their enjoyment of life's benefits. Toynbee argues that the gas guzzling 4×4 owner "will have to buy a quota" – but the critics might describe matters differently, perhaps saying: "The rich gas-guzzling 4×4 owner is given a chance to buy access to the road and to exclude the poorer motorist from the highway." Such critics might observe that when Robin Hood took from the rich and gave to the poor he did not, in return, limit the poor's entitlements to exploit the resources of Sherwood Forest.

The "free lunch" conception of emissions trading breaks down on scrutiny. Enough was said in Section 2 above to indicate that emissions trading, at least in some forms and contexts, is subject to question on a number of fronts – notably regarding its objectives, cost-effectiveness, fairness, transparency, and accountability. Nor is it convincing to argue that trading is a device that stands to be judged by the canons of market accountability. The problem with emissions trading is that, even if trading is healthy, the market may not deliver an acceptable set of outcomes by means of acceptable processes.

The second assumption consistent with a shift toward thinner expectations of legitimacy is the realpolitik notion that this mechanism is acceptable because it can be implemented. Emissions trading may be viewed as "regulation lite" by critics because it frequently involves allocations that are designed not to "frighten the horses" of the incumbents. That "lite" quality, however, may be welcomed by many governments on the grounds that, at least on the world stage, we face global warming issues of such urgency that the best regulatory method for controlling greenhouse gases is the one that has the best chance of implementation (Butzengeiger & Michaelowa 2004, p. 116). They might add that, since 1992, prior attempts to introduce carbon taxes had failed and no other policy instrument was delivering emission reductions.

Emissions trading, moreover, can be said to possess three valuable attributes that make it a specially attractive proposition when acceptability is at issue. First, it holds out the prospect of a precise goal (the overall emissions cap) combined with a procedural framework, and, as such, it provides an identifiable aim that different parties or countries

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can negotiate and agree on. Its implementability is its strength. Second, it allows a nettle to be grasped now while deferring the resolution of difficult issues to a later date (similar to the situation in which the EU ETS deferred many issues by moving them into the entitlements allocation processes). Third, emissions trading both appeases powerful players (by accepting their incumbent positions as givens) and it also offers short-term compensation to less well placed parties. All of these properties have been noted by proponents of emissions trading. Gordon Brown welcomed the Stern Review by emphasizing that multilateral action on the environment had to be taken with urgency and that devices such as the EU ETS were of key importance in "making the economic opportunities of a climate friendly policy real and tangible" (*The Guardian* 30 October 2006). These comments echoed Stern's assertion that it was essential "to create a shared international vision of long term goals and to build international frameworks" (Stern 2007, p. xviii). What was on offer, then, was not only a focused target but economic opportunities for business and the instigation of some action for green pressure groups. Thus Friends of the Earth dubbed the EU ETS a "potentially huge step forward in the race to tackle climate change" (Friends of the Earth 2004). Left out of account in such reactions, though, are the forgone opportunities suffered by sellers of allocations.

Where, though, does this leave any analysis of regulatory quality? In the global warming context, at least, emissions trading no longer appears to have to satisfy the Five Principles in order to constitute "good" regulation – emissions trading seems to qualify as "good" if it meets two tests. First, it has the potential to address an issue of catastrophic risk. In other words, anything that works in an emergency is acceptable. It can be implemented, and notions of accountability and fairness are demoted in importance because effectiveness in achieving objectives is paramount; or because there are assumed to be no losers within trading systems; or because public notions of accountability are deemed to be adequately replaced with accountability through markets. Second, a system of regulation is held to be good because it is cost-effective – that is, it provides a desired solution at lowest cost and it respects the rights of incumbents.

The difficulty with these defenses is that, as already noted, trading in itself does not do much at all to address the catastrophic risks facing the world; it merely offers a low-cost option for action. There is, moreover, no free lunch. There can be significant losers within trading mechanisms, namely those who start off with fewer chips in life's poker game. Accountability to the market may exist but that market is skewed toward incumbents and powerful players. A final difficulty with emissions trading is that even if we were prepared to tolerate its weakness of accountability and fairness on the grounds of an anticipated response to a catastrophic risk, we cannot apply the same reasoning to non-catastrophic risk, such as the risks of traffic congestion. This is especially the case where alternative regulatory instruments might achieve similar objectives.

At this point it might be contended: Why not opt for emissions trading because the available alternative modes of controlling pollution – such as commands or taxes – are equally fraught with dangers (regarding, for example, manipulation by powerful interests) but emissions trading at least offers cost-effectiveness and ease of implementation? On this point it must be emphasized again that much depends on how emissions trading is put into effect; on which version of trading is applied in a given context; and on the potential of steps that can be taken to reduce the negative effects of trading. Within these

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bounds of generalization, though, the answer to the "all systems are difficult" point is twofold. First, there are reasons, as indicated, to think that emissions trading, at least in certain circumstances, aggravates some of the problems commonly associated with traditional forms of command regulation. For instance, it can involve particular incentives to manipulate, special incumbent advantages, and notable difficulties of accountability and transparency. In addition, it forgoes across-the-board standards in favour of less fair mechanisms that tend to reward past polluters. Second, it can be replied that if we are to choose emissions trading, we should not do so on a "free lunch" basis or because "all controls are fraught" but because we place primacy on its capacity for overcoming the opposition of the economically powerful. This leaves a final issue: how such regulation is to be justified to the broad public.

3.2. Justification in the era of emissions trading

As far as theories of regulatory legitimization and accountability are concerned, there is an important message to be gleaned from the above discussion. It is that there may be material limitations on the extent to which "market" mechanisms can be deployed constructively alongside traditional "democratic" mechanisms of accountability, transparency, or legitimization. This suggests, in turn, that a degree of caution may be appropriate regarding the possibility of developing coherent regimes of accountability in complex regulatory systems in which the state is not the sole locus of authority but where control functions are spread across a variety of state and non-state actors (Black 2001).

Such caution contrasts with the optimistic vision of the "redundancy" model of accountability. This model suggests that, in complex systems of modern government, control and policy functions are often spread across institutions and processes of different kinds, that conflicts and tensions exist within the complex accountability webs that apply within a regulated domain, but that "the objective should not be to iron out conflict, but to exploit it to hold regimes in appropriate tension" (Scott 2000). The approach is optimistic insofar as it stresses that the various accountability systems that operate concurrently in an area "have the character of a complex system of checks and balances" which have the potential for being harnessed into an effective accountability system "even as public power continues to be exercised in more fragmented ways" (Scott 2000, p. 55; Stirton & Lodge 2001).

Experience with emissions trading suggests that there are a number of reasons why the sanguineness of the redundancy model may be inappropriate in the post-emissions trading era. This, in short, is because emissions trading systems can raise accountability and legitimization challenges to unprecedented levels. Even before the rise of emissions trading, those challenges were rendered severe by the development of "decentered" regulatory systems. In decentered systems, the mechanisms and processes of holding to account involve not merely variations in institutions, procedures, discourses, systems, and expectations across actors or constituencies, but also the potential for destructive, rather than harmonious, interactions between mechanisms. Such difficulties arise, for instance, when certain types of accountability relationships are constructed between parties in ways that are in tension with other relationships. This can occur because different processes are used but also where different frames of meanings, values, and objectives are involved (Black 1998, 2002).

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In the case of emissions trading, however, such difficulties rise to new levels due to three significant factors. First, the assumptions that underpin some accountability relationships within trading are inconsistent with other modes of control. Thus, the assumption that emissions trading processes involve no losers is compatible with an efficiency based reliance on market controls, but it does not sit easily with the notion that less affluent citizens need democratically based protections from the distributional consequences of market transactions.

Second, within emissions trading there are control systems that operate with inconsistent core tenets. Thus, the idea that the market will allocate emissions abatements in an uncontentious manner is at odds with beliefs that the distribution of market rights may need to be adjusted in order to further social justice. The view that distributional decisions can be overlaid on market mechanisms in a transparent way is liable to be heavily disputed by those who think that such efforts underestimate market power and will, at best, involve tinkering in the face of the overwhelming need to preserve the workings of the market.

Third, there are incompatibilities within the system regarding the very needs for legitimacy and accountability. Emissions trading is not a system in which "market" and "democratic" checks and balances can be brought into line with any ease. Normally an observer might view a "market" mechanism as bringing accountability to consumers, shareholders, and other stakeholders, and might see "democratic" mechanisms as ensuring accountability to citizens and participants. In emissions trading, however, the "market" is self-regarding and "closed" in nature so that (without strong corrective instruments) there is not even an effective regime of control by consumers, shareholders, or others. Governments which institute emissions trading systems allocate permits for trading between polluters, not between polluters and consumers. Such governments, accordingly, are involved in a process that relinquishes their own role as holders to account and reasserts their role only when they supplement trading with further controls or reset the emissions caps or baselines. They do not give up this regulatory role in favor of holding to account by consumers but set the market free – often on the ground that this is justifiable because the need for some action to combat emissions trumps any need for legitimization beyond the group of potential compliers. Trading systems, accordingly, can constitute "accountability black holes" and, as such, cannot be harnessed alongside other accountability mechanisms in a coherent legitimizing mesh. Regarding legitimization, trading might be said to generate mush, not mesh.

The optimistic version of the redundancy model stresses the potential for harmony and coherence in concurrent accountability mechanisms. The pessimistic vision suggests that redundancy theory may offer a valuable perspective on control regimes but that there are good reasons for thinking that mixtures of accountability systems will sometimes produce confusions, uncertainties, injustices, and democratic deficits. This is liable to occur when there are inconsistencies in the assumptions that form the basis for controls, when there are incompatibilities of relevant values, discourses and visions of accountability, and when there are variations in the accountability objectives of different systems (Black 2001; Baldwin & Black 2008). The challenge for public lawyers and the designers of regulatory systems may be to make appropriate interventions in order to make complex networks of accountability work (Scott 2000). The emissions trading experience, however, suggests that such a challenge may be Herculean when the above three factors are encountered.

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Conclusions

Sweeping conclusions regarding emissions trading should be offered and treated with caution for reasons already given. In divergent contexts, trading regimes display very different potential – both in absolute terms and in comparison with other regulatory instruments. Thus, with relatively small numbers of well-resourced emitters and continuous emissions monitoring devices – as in the US Acid Rain Program – the problems faced do not occupy the same scale as those encountered in a decentralized and international greenhouse gas system with large numbers of varyingly resourced emitters and daunting challenges regarding monitoring and enforcement. It is often difficult, moreover, to pinpoint the extent to which observed difficulties are inherent in the trading device or can be explained as teething troubles (as with the EU ETS) or as implementation failures (that might or might not afflict other regulatory instruments). Some problems (e.g. of accountability and transparency) may, furthermore, be pointed out in trading regimes but may be readily addressable through supplementary controls. That said, it is to be expected that the difficulties associated with emissions trading regimes will tend to be the greater when they involve such factors as: high numbers of regulated organizations or regulators; decentralized regimes; cross-jurisdictional applications; and high variations in resources and competencies across regulated enterprises or regulators. Further such factors include: complexities in the allowances traded; inequalities of wealth or pollution records that raise contentious redistributive questions; serious enforcement and monitoring issues; stringent emissions targets; and high levels of incumbency power.

In the light of the above caveats and the prior discussion, how is the rise of emissions trading to be responded to? There are, perhaps, two ways forward. One is to accept that "anything goes" – that, when faced with catastrophic risks, it is churlish to complain about legitimacy, accountability, or inherent biases in markets. The other way is to resist the optimistic view of redundancy and to reassert democratic values.

The latter approach might well involve working toward ways of controlling the negative aspects of trading mechanisms (such as their opacity) while holding, first, that arguments predicated on the confronting of catastrophic risks should not be applied in the case of non-catastrophic risks, and, second, that it is especially when faced with potential catastrophes – and when the most dramatic decisions are taken – that we have to be most wary of the optimistic (or "romantic") vision of redundancy and most reluctant to settle for confused regimes of control and legitimation.

The modern use of emissions trading has produced a model of regulation that is "lite" in its inherent incumbent friendliness and that is resonant with assumptions that are highly contentious. If we are to be more positive about the use of emissions trading, we might say that it has produced a level of philosophical confusion about regulatory virtue that is patent enough to create two important opportunities. The opportunity for governments (and for commentators) is to reformulate their conceptions of good regulation and to lay to rest any notions that regulation through trading is wholly acceptable because it involves no losers or because, in desperate circumstances, we have no choice. The opportunity for regulatory theorists is one that should not be spurned. It is to confront the difficulties of combining different systems of legitimization rather than to take refuge in all too comfortable beliefs in cumulative checks and balances.

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Notes

1 The characteristics are: scope; cap; commodity traded; distribution of permits; trading ratio; banking; monitoring; and environmental benefit.

2 Open Europe notes that Member States handed out permits for 1,829 million tonnes of CO2 in 2005 while emissions were only 1,785 million tonnes. In 2006 carbon emissions grew by 1.25 percent on 2005 – see, "Carbon Emissions Rise", *Financial Times* 30 March 2007. Over-allocation of allowances has occurred in other regimes – such as the Los Angeles Regional Clean Air Incentives Market (RECLAIM) (Tietenberg 2006).

3 This is not to deny that in other circumstances the trading process can help to flush out information – see Sanchez and Katz (2002) and Penderson (2003).

4 On "benchmarking" as an alternative to allocations based on past emissions, see Ellerman and Buchner (2007), pp. 76–78.

5 Mumma has argued that many developing countries lack the financial, technical and human resources necessary to allow them to negotiate equally with developed nations on emissions trading issues or to evaluate emissions trading programs thoroughly enough to judge where their longer-term interests lie (Mumma 2001).

6 On the discourse of "ecological modernization" and optimism regarding the potential to meet economic, social, and environmental objectives in a manner that does not threaten market principles, see generally Hajer (1995), and Young (2000). On markets and regulation, see Braithwaite (2005).

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